

EXAM 7 – SPRING 2011

3. (4 points)

One implicit assumption underlying the chain ladder loss development method is independence of the accident years. Given the following loss development triangle and link ratios:

Accident Year	<u>Reported Losses (\$000)</u>				
	12 Months	24 Months	36 Months	48 Months	60 Months
2006	3,100	4,185	4,813	5,053	5,104
2007	3,050	3,660	4,282	4,368	
2008	2,800	3,640	4,077		
2009	2,500	3,125			
2010	2,357				

	<u>Age-to-Age Loss Development Factors</u>			
	<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>
2006	1.35	1.15	1.05	1.01
2007	1.20	1.17	1.02	
2008	1.30	1.12		
2009	1.25			

a. (3 points)

Using a 95% confidence interval, test the null-hypothesis that this triangle does not have significant calendar year effects.

b. (1 point)

Describe two other implicit assumptions underlying the chain ladder loss development method.

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Sample Solutions
Exam 07, 2011, Question 3

Solution 1

a)

95% CI: statistic = 1.96

$$\begin{array}{rcl} \text{Median} = & \frac{12-24}{(1.25+1.3)/2} & \frac{24-36}{1.15} & \frac{36-48}{(1.02+1.05)/2} \\ = & 1.275 & 1.15 & 1.035 \end{array}$$

For development factors greater than median, its given "L", otherwise, given "S"

AY	12 - 24	24 - 36	36 - 48
06	L	×	L
07	S	L	S
08	L	S	
09	S		

$$E(Z_j) = n/2 - \binom{n-1}{m} \times (n/2^n)$$

$$\text{Var}(Z_j) = n \times (n-1)/4 - \binom{n-1}{m} \times (n \times (n-1)/2^n) + E(Z) - E(Z)^2$$

$$n = S + L$$

$$m = (n-1)/2$$

$$Z = \min(S, L)$$

Let j = Diagonal

J	N	m	Z	E(Z)	Var(Z)
1	1	0	0	0	0
2	1	0	0	0	0
3	3	1	0	0.75	0.1875
<u>4</u>	<u>3</u>	<u>1</u>	<u>0</u>	<u>0.75</u>	<u>0.1875</u>
			0	1.50	0.375

Null hypothesis = that there are no significant calendar year effects

Confidence interval: $1.5 \pm 1.96 \times \sqrt{0.375} \Rightarrow (0.275, 2.725)$

Since $Z=0$ is not contained in the interval, you reject the null hypothesis and suggest that there are significant calendar year effects

b)

Assumption 1 - the future incremental expected loss is proportional to the losses cumulative to date

Assumption 2 - the variance of the next incremental loss is a function of the age and cumulative losses to date